



Implicit-Explicit Runge Kutta Methods in Design Optimization with Applications to Heat Transfer

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Outline

- 1: Overview.
- 2: Background and Motivation.
- 2: Automatic Differentiation of IMEX Schemes.
- 3: Optimization Examples.



Presentation Overview

- ▶ This presentation will illustrate a method for exploiting the structure of topology optimization problems with mixed multiphysics constraints.
- ▶ We construct a unified framework for adjoint-based design approach using IMEX-RK schemes for forward- and backward-time integration.
- ▶ Implement a routine which automatically differentiates IMEX-RK methods.
- ▶ Present large heat transfer topology optimization experiments in MFEM demonstrate to the efficacy of our framework in the high-performance computing (HPC) context.

Topology Optimization

- ▶ We utilize a density based approach to topology optimization. Consider an open bounded Lipschitz domain $\Omega \in \mathbb{R}^n$. Let X be a Hilbert space of functions over Ω .
- ▶ Let u be a function which describes the physical state. Let R be the residual of the governing physics equations. We seek a density field ρ which solves:

$$\begin{aligned} \min_{\rho \in X} \quad & J(\rho, u) \\ \text{s.t.} \quad & R(\rho, u) = 0 \end{aligned} \tag{1}$$

- ▶ $\rho = 1$ corresponds to a void region, while $\rho = 0$ is a material region.



Motivation for IMEX-RK Schemes in Design

- ▶ Often, R will represent a set of coupled mixed-multiphysics PDEs.
- ▶ Computational cost can be greatly reduced when we exploit the structure of the underlying PDEs.
- ▶ For time dependent PDEs, Implicit-Explicit Runge-Kutta (IMEX-RK) time integration has proven highly successful, especially for fluid dynamics and heat transfer problems.

Intro to IMEX-RK Schemes

- **IMEX Scheme.** Let $t \in [0, T]$, $T > 0$. Consider a time-dependent ODE of the form:

$$\frac{du}{dt} = f_1(u, t) + f_2(u, t). \quad (2)$$

Let f_1 be difficult to integrate implicitly (possibly non-linear). Let f_2 be stiff. An IMEX scheme applies an explicit integration scheme to f_1 and implicit to f_2 .

- **Runge-Kutta IMEX.** In an IMEX-RK scheme, we use an s -stage explicit RK scheme to integrate f_1 and an s -stage implicit to integrate f_2 . The explicit scheme is given by a butcher table with coefficients $\hat{A}, \hat{b}, \hat{c}$. The implicit scheme is given by a butcher table with coefficients A, b, c .

Intro to IMEX-RK Schemes

Let i be the time step index, let j be the RK stage index. The resulting IMEX-RK Scheme is:

$$\begin{aligned}
 u_i &= u_{i-1} + \Delta t \sum_{j=1}^s \hat{b}_j \hat{K}_{i-1,j} + \Delta t \sum_{j=1}^s b_j K_{i-1,j} \\
 \hat{K}_{i-1,j} &= f_1(U_{i-1,j-1}) \\
 K_{i-1,j} &= f_2(U_{i-1,j}) \\
 U_{i-1,j} &= u_{i-1} + \Delta t \sum_{l=1}^j \hat{a}_{j+1,l} \hat{K}_{i-1,l} + \Delta t \sum_{l=1}^s a_{j,l} K_{i-1,l}.
 \end{aligned} \tag{3}$$

Automatic Differentiation

- From (1), we need to compute:

$$\lambda^T \frac{\partial R}{\partial \mathbf{u}} = -\nabla_{\mathbf{u}} J$$

and

$$\lambda^T \frac{\partial R}{\partial \rho}(\mathbf{u}) = -\nabla_{\rho} J$$

- To do this, we need a way to efficiently differentiate (3). **For high order schemes, it's a mess.**
- U_j in (3) depends on all of the previous stages.
- So long as the implicit scheme is DIRK, we have implemented in MFEM a general method for differentiation for an arbitrary number of stages and order.

Automatic Differentiation - Adjoint

- Multiply by adjoint variable λ_i and differentiating with respect to u_i gives:

$$\begin{aligned}\lambda_{i-1}^T &= \lambda_i^T + \Delta t \sum_{j=1}^s \hat{b}_j \lambda_i^T \frac{\partial \hat{K}_j}{\partial u_{i-1}} + \Delta t \sum_{j=1}^s b_j \lambda_i^T \frac{\partial K_j}{\partial u_{i-1}} \\ \frac{\partial \hat{K}_j}{\partial u_{i-1}} &= \frac{\partial f_1}{\partial U_{j-1}} \left[\sum_{l=1}^j \frac{\partial U_{j-1}}{\partial U_{j-l}} \frac{dU_{j-l}}{du_{i-1}} \right] \\ \frac{\partial K_j}{\partial u_{i-1}} &= \frac{\partial f_2}{\partial U_{j-1}} \left[\sum_{l=1}^j \frac{\partial U_{j-1}}{\partial U_{j-l}} \frac{dU_{j-l}}{du_{i-1}} \right]\end{aligned}\tag{4}$$

- Here, $\frac{d}{du_{i-1}}$ represents the total derivative. There is a recursive structure which, for arbitrary DIRK IMEX schemes, makes it tedious to derive by hand but simple to code.

Automatic Differentiation - Design Gradients

- Multiply by adjoint variable λ_i and differentiating with respect to ρ gives:

$$\begin{aligned}
 0 &= \Delta t \sum_{j=1}^s \hat{b}_j \lambda_i^T \frac{\partial \hat{K}_j}{\partial \rho} u + \Delta t \sum_{j=1}^s b_j \lambda_i^T \frac{\partial K_j}{\partial \rho} u \\
 \frac{\partial \hat{K}_j}{\partial \rho} &= \frac{\partial f_1}{\partial \rho} + \frac{\partial f_1}{\partial U_{j-1}} \left[\sum_{l=1}^j \frac{\partial U_{j-1}}{\partial U_{j-l}} \frac{dU_{j-l}}{d\rho} \right] \\
 \frac{\partial K_j}{\partial \rho} &= \frac{\partial f_2}{\partial \rho} + \frac{\partial f_2}{\partial U_{j-1}} \left[\sum_{l=1}^j \frac{\partial U_{j-1}}{\partial U_{j-l}} \frac{dU_{j-l}}{d\rho} \right]
 \end{aligned} \tag{5}$$

- **Implementation.** One of our main contributions is we have implemented a routine in MFEM to automatically compute these derivatives - for any DIRK IMEX-RK scheme.

Examples!

- ▶ IMEXcited to show you some optimization examples using IMEX-RK automatic differentiation....

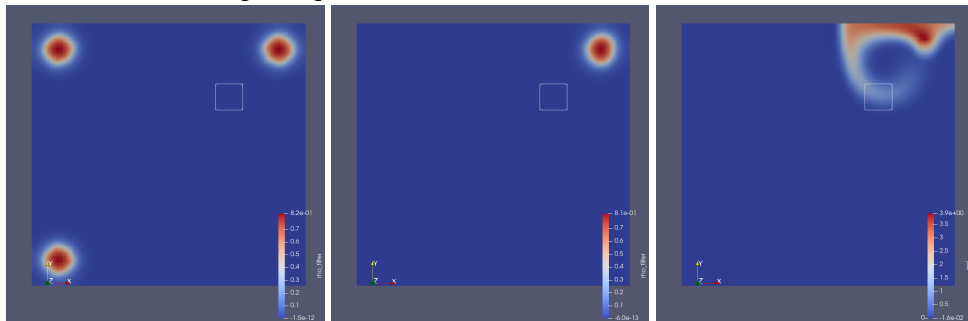
Example 1 - Injection Optimization

- We seek to optimize flux through a sub-region of the domain. Let $\Omega \subset \mathbb{R}^2$. Let $B \subset \Omega$ be a target region. Let $A \subset \Omega$ be the "flux" region. Let $t \in [0, T]$. We seek to solve:

$$\begin{aligned}
 \min_{\rho} \quad & \int_B \theta(x, T)^2 dx \\
 \text{s.t.} \quad & \frac{\partial \theta}{\partial t} = -\mathbf{v} \cdot \nabla \theta + \nabla \cdot (k \nabla \theta) + f(\rho) \text{ in } \Omega \\
 & f = g(x, \rho) \text{ in } A \\
 & f = 0 \text{ in } \Omega - A \\
 & \int_A f(\rho) dx \leq d
 \end{aligned} \tag{6}$$

Example 1 - Injection Optimization

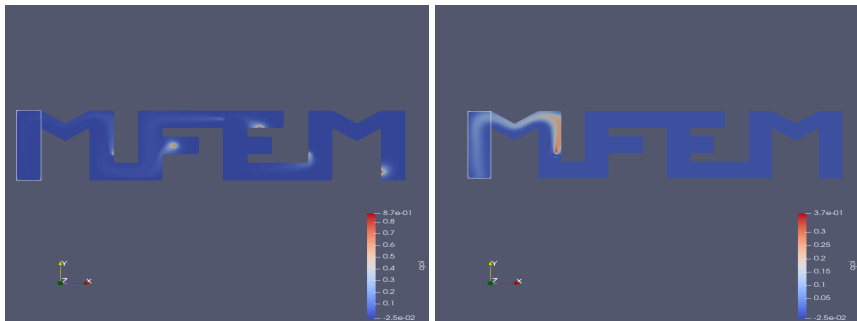
- ▶ **Left:** Initial Flux Density Field. **Middle:** Final Flux Density. **Right:** θ at final time.
- ▶ White box is the target region B .



- ▶ Experiments performed with 2nd order IMEX-RK schemes.

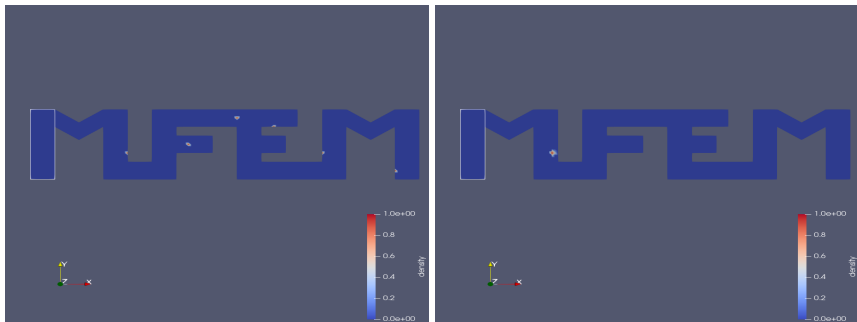
Example 1 - Injection Optimization

- **Left:** Unoptimized injections at final time. **right:** Optimized injections at final time.



Example 1 - Injection Optimization

- **Left:** Unoptimized flux densities. **right:** Optimized fluxes.



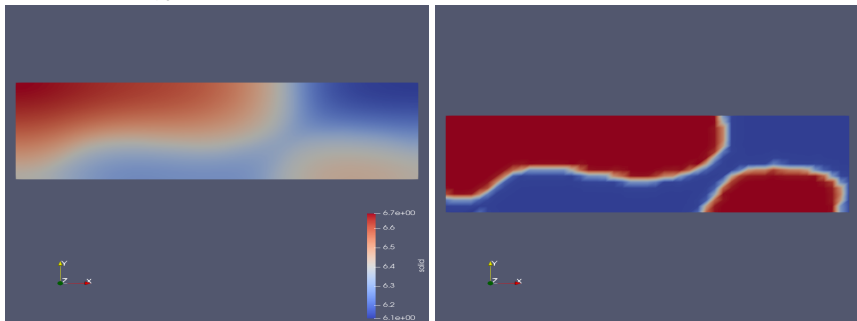
Example 2 - Heat Transfer Topology Optimization

- We seek to optimize the design of heat sinks under forced Convection. We borrow the pseudo-3D framework from [1]. Let $\Omega_f, \Omega_s \subset \mathbb{R}^2$. Let Ω_s be the base plate of the heat sink. Let Ω_f be a cross section of the fluid region and heat sink fins. Let $t \in [0, T]$.

$$\begin{aligned}
 \min_{\rho \in X} \quad & \int_{\Omega_s} \theta_s(x, T)^2 dx \\
 \text{s.t.} \quad & \frac{\partial \theta_f}{\partial t} = -\mathbf{v} \cdot \nabla \theta_f + \nabla \cdot (c(\rho) \nabla \theta_f) + \mathcal{I}_h(\rho)(\theta_f - \theta_s) \text{ in } \Omega_f \\
 & \mathbf{0} = -\nabla p + \epsilon_f \Delta \mathbf{v} + \alpha(\rho) \mathbf{v} \text{ in } \Omega_f \\
 & \frac{\partial \theta_s}{\partial t} = \nabla \cdot (c_s \nabla \theta_s) - \mathcal{I}_h(\rho)(\theta_f - \theta_s) + Q(x) \text{ in } \Omega_s
 \end{aligned} \tag{7}$$

Example 2 - Heat Transfer Topology Optimization

- **Left:** Heat field of the base plate at final time $T = 1.0$. **Right:** Filtered optimized topology.



Questions?



References

-  J. H. K. HAERTEL, K. ENGELBRECHT, B. S. LAZAROV, AND O. SIGMUND, *Topology optimization of a pseudo 3d thermofluid heat sink model*, International Journal of Heat and Mass Transfer, 121 (2018), pp. 1073–1088.



